IDEATION AND BUSINESS MODEL DESIGN / 24 HOUR DEVELOPMENT SPRINT

Prototype Readiness Report

Complete after the 11-day build. Report what actually works; enter concise evidence. One team, one page.

PROJECT / PROTOTYPE
CropDoc AI: on-phone AI crop disease diagnosis with Telugu advice for Hyderabad-region farmers (paddy, cotton, maize, chilli, tomato)

TEAM / MEMBERS
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COURSE / FACULTY
Ideation and Business Model Design / Prof. Geeta Tripathi, Woxsen University

SPRINT DATES / SUBMISSION DATE
01/11 – 02/11/2026 (submitted 02/11/2026)

01 Baseline & 24-hour goal

What worked at the start?	Business model canvas deck (v1, 14 Oct 2026) and a click-through UI mock-up with simulated results. No trained model; disease answers were example outputs only.
End-to-end use case goal	A farmer picks the crop, photographs a leaf on any phone, and within 2 s gets the disease name, a confidence score, Telugu text and voice advice and the spray dose, with the AI running on the phone.

02 Feature commitment and actual delivery

Maximum 2 core features + 1 stretch feature

Priority	Feature + acceptance / success criterion	Before?	At 24h: status + evidence
CORE 1	On-phone AI diagnosis (MobileNetV3, 36 classes, 5 crops). Success: ≥90% accuracy on unseen test photos and <2 s per photo.	No	Done. 93.6% correct on 1,833 held-out photos; 60–180 ms per photo in the browser (ONNX Runtime Web).
CORE 2	Telugu advice and safety gate: Telugu/English advice, voice and spray-dose calculator for all 36 classes; below 70% confidence the app says “ask an expert”. Success: 36/36 classes covered; confident answers ≥95% correct.	No (mock text for 4 classes only)	Done. 36/36 classes with doses from PJTAU/ICAR sources; at ≥70% confidence the AI was 97.7% correct (88% of test photos).
STRETCH	Farmer feedback and expert escalation: “Was this right?” capture, WhatsApp ask-expert message, CSV export. Success: feedback saved with every scan and exportable.	No	Partial. Works on the phone (local storage) and CSV copy works; no server yet, so feedback is not collected centrally.

03 What was built & how it will be demonstrated

Our build / external tools	Our build: data pipeline (18,359 images, 36 classes), MobileNetV3 fine-tuning, held-out evaluation, ONNX export, web app with Telugu advice, voice, dose calculator and history. External: ImageNet weights (timm), PyTorch, ONNX Runtime Web, public datasets (Paddy Doctor, SAR-CLD-2024 cotton, PlantVillage tomato, maize, chilli), Claude AI assistant for coding and research, Gamma for slides.
Live demo steps (1 → 2 → 3)	1) Open the app link and choose a crop (e.g. Paddy) → 2) take or upload a leaf photo, or tap an example test photo → 3) show the diagnosis and confidence, play Telugu voice, set acres in the dose calculator, answer “Was this right?” and open the WhatsApp expert message.

04 Verification

Show two real tests, not just screenshots

Test / input	Expected output	Observed result / measurement	Pass / fail
1,833 held-out test photos never used in training; crop selected first, as in the app	≥90% correct	93.6% overall (tomato 98.4%, maize 95.9%, cotton 95.3%, paddy 87.5%, chilli 80.0%); 97.7% when confidence ≥70%	Pass
10 example test photos run in the live hosted app in Chrome	Correct disease in <2 s; unsure cases flagged, not stated as fact	8/10 correct at 94–99% confidence; 2 flagged as “Not sure” (30%) and “Possible” (52%); 59–179 ms per photo	Pass

05 Readiness & TRL mapping

Use observed technical evidence to choose both values

Classroom readiness (R0–R4) mapped to indicative technology maturity (TRL 1–6):

R0 | TRL 1–2 **R1 | TRL 2–3** **R2 | TRL 3–4** **R3 | TRL 4** **R4 | TRL 5–6**

Concept and basic principles	Components tried; critical-function PoC if shown	Core function works; lab integration if verified	Integrated prototype, repeatable lab tests	Relevant-environment validation / demo

TRL rule: TRL 3 = critical-function proof; 4 = lab validation; 5 = relevant-environment validation; 6 = prototype demo there. Feedback alone does not establish TRL 5.

BEFORE sprint: R1 / TRL 2	Evidence: canvas deck and UI mock-up with simulated results; no trained model and no measured accuracy.
AFTER 24h: R3 / TRL 4	Measured proof: integrated app and model; repeatable lab tests on 1,833 held-out photos (93.6%) and automated browser runs. Not TRL 5: no validation in Telangana fields yet.

06 Limitations, next action & proof

BIGGEST LIMITATION / FAILING FEATURE
Not yet tested on Telangana farm photos or with farmers. Paddy 87.5% and chilli 80.0% (chilli leaf-curl complex only 50%). Advice not yet expert-reviewed. First load needs about 31 MB of internet. No “not a leaf” class.

NEXT 7-DAY IMPROVEMENT
KVK Ranga Reddy / PJTAU review of all 36 advice entries; field test with 10–20 farmers near Hyderabad; collect 300+ labelled local photos and report field accuracy.

EVIDENCE LINK (CODE / PHOTOS / LOGS / VIDEO)
App: https://claude.ai/artifact/6pThVz97PXtL7yQNWP2QNQ · Report: https://claude.ai/artifact/3SiteA96rasUVXh3BNZtkM · Deck: https://gamma.app/docs/CropDoc-AI-bri75pz99ppkrrx · Code + model: cropdoc-ai-v2.zip

TEAM LEAD / SUBMISSION CONFIRMATION
Submitted by Arjun Dosapati (24WU0102184) for the team, 02/11/2026

07 User validation & comments

If any — report actual user feedback

VALIDATION STATUS
Not conducted

USER TYPE / NUMBER TESTED
No external users yet; build team only. Planned: 10–20 farmers near Hyderabad and 1 KVK expert.

TASK TRIED + USER RESULT
Internal test only: diagnose 10 example photos end to end. Worked; 8/10 correct, 2 flagged as unsure.

USER COMMENTS + PROPOSED FIX
No real user comments yet. Fixes from internal test: replace AI-generated deck images with real app photos; add recorded Telugu audio for phones without a Telugu voice.

DECLARATION We distinguish implemented, simulated and planned features; identify external tools; and report only user tests actually conducted.

TRL mapping guide based on NASA technology readiness definitions (TRL 1–9). R0–R4 is a classroom scale, not a certified TRL assessment; TRL 7–9 require later operational-system evidence.

NOTE: Sprints determine the time period you are achieving a functionality